Perioperative Fluid Therapy and Goal-Directed Haemodynamic Optimization

ESICM Consensus & Enhanced Recovery After Surgery (ERAS) Protocols

1. Fluid Types and Selection

CRYSTALLOIDS (isotonic):

- Normal Saline (0.9% NaCl): risk of hyperchloremic metabolic acidosis with large volumes

- Balanced crystalloids (Lactated Ringer's, Plasmalyte): preferred for high-volume therapy;

lower chloride load; supports physiologic acid-base balance

- Dose: 1–2 mL/kg/hr as maintenance; additional boluses guided by dynamic response

COLLOIDS:

- Albumin (4% or 5%): plasma expander; preferred in critically ill or hypoalbuminaemic patients

- Hydroxyethyl Starch (HES): associated with AKI and coagulopathy in critically ill;

use restricted by EMA since 2023; avoid in renal impairment

- Gelatin-based colloids: widely used in Europe; limited evidence vs albumin

BLOOD PRODUCTS:

- Packed RBCs: trigger Hb < 7 g/dL general; < 8 g/dL cardiac patients

- FFP: coagulopathy with ratio 1:1:1 in massive hemorrhage protocols

2. Goal-Directed Fluid Therapy (GDFT)

GDFT uses real-time hemodynamic monitoring to guide fluid delivery beyond static protocols.

KEY DYNAMIC VARIABLES:

- Pulse Pressure Variation (PPV): >13% predicts fluid responsiveness in ventilated patients

- Stroke Volume Variation (SVV): >10–12% indicates preload dependence

- Passive Leg Raise (PLR): a 10% increase in cardiac output confirms fluid responsiveness

without administering fluid; useful in spontaneously breathing patients

CARDIAC OUTPUT MONITORING DEVICES:

- Oesophageal Doppler (CardioQ): guides fluid in major abdominal surgery

- FloTrac/Vigileo: pulse contour analysis; SVV calculation

- LiDCO, PulseCO: lithium dilution calibration; arterial waveform CO

- Transoesophageal Echo (TOE/TEE): gold standard; not continuous

GDFT CLINICAL IMPACT:

- Reduces post-operative complications by 35% in high-risk surgery (meta-analysis, 2019)

- Reduces hospital LOS by 1.2 days in colorectal surgery

- Reduces IOH incidence by preventing relative hypovolaemia

3. Fluid Pre-loading and Co-loading

SPINAL ANESTHESIA — PRE-LOADING:

- Crystalloid pre-load (1–1.5 L) before spinal induction reduces IOH incidence but effect

is transient due to rapid redistribution; less effective than co-loading or vasopressors.

- Colloid pre-load (500 mL albumin or HES): more sustained intravascular effect;

reduces IOH incidence by ~50% vs no pre-load (Mercier et al., 2010).

CO-LOADING (simultaneous with spinal induction):

- More effective than pre-loading alone; maintains circulating volume as sympathectomy develops

- Crystalloid co-load 500–1000 mL at induction; preferred in obstetric spinal anesthesia

- Combined crystalloid co-load + prophylactic phenylephrine infusion: gold standard

for elective cesarean section (Dennis et al., 2012; Carvalho et al., 2017).

RESTRICTED vs LIBERAL FLUID THERAPY:

- Perioperative restrictive fluid therapy reduces wound complications and pulmonary edema

- Excessive fluid (>3 L crystalloid) worsens gut function, increases abdominal compartment

syndrome risk, and is associated with anastomotic leak in colorectal surgery (RELIEF trial)

- Individualized GDFT is superior to either fixed protocol